

Take home exam: Causal Machine Learning

Only for students of the Master of Mathematics

This take home exam must be made individually. The accompanying materials around M-estimators and non-parametric inference will not be examined further during the final exam. The deadline is the day of the exam, but I encourage you to submit your solutions earlier so that you obtain intermediate feedback.

1. (Exercise on M-estimation) Suppose that we are interested in the effect of a binary exposure A (coded 0 or 1) on a continuous outcome, and suppose that a scalar measurement L suffices to adjust for confounding. Suppose that

$$P(A = 1|L) = \frac{\exp(\alpha_0 + \alpha_1 L)}{1 + \exp(\alpha_0 + \alpha_1 L)} \quad (1)$$

$$E(Y|A, L) = \beta_0 + \beta_1 A + \beta_2 L + \beta_3 L^2 \quad (2)$$

Suppose first that the propensity score (and thus α_0 and α_1) is known, and that we conduct the following analysis in R:

```
model <- lm(y ~ a + ps),
```

where **ps** is the vector $\left(\frac{1}{1 + \exp(-\alpha_0 - \alpha_1 L_i)} \right)_{i \in [n]}$. Here, the last line produces an ordinary least squares (OLS) estimate

$$\left(\sum_{i=1}^n X_i(\alpha) X_i(\alpha)^\top \right)^{-1} \sum_{i=1}^n X_i(\alpha) Y_i, \quad X_i(\alpha) := \left(1, A_i, \frac{1}{1 + \exp(-\alpha_0 - \alpha_1 L_i)} \right)^\top.$$

- (a) Show that the influence function of the obtained OLS estimator of the coefficient of **a** in this final analysis equals

$$\{A - P(A = 1|L)\} \{Y_i - \gamma_0 - \gamma_1 A_i - \gamma_2 P(A = 1|L)\}$$

up to a proportionality constant, for some value $(\gamma_0, \gamma_1, \gamma_2)'$. *Hint. Note that there are nuisance parameters, namely the intercept and the slope of the propensity score in the outcome model. Account for their uncertainty like we did in the section on ‘Nuisance parameters’.*

- (b) Show that this influence function has mean zero under model (2) if and only if $\gamma_1 = \beta_1$. What can we conclude from this?

Suppose next that α_0 and α_1 are unknown, and therefore estimated using maximum likelihood. We thus instead conduct the following analysis in R:

```
ps <- glm(a ~ 1, family=binomial)$fitted
model <- lm(y ~ a + ps),
```

where `ps` is now the vector $\left(\frac{1}{1+\exp(-\hat{\alpha}_0-\hat{\alpha}_1 L_i)}\right)_{i \in [n]}$ with $\hat{\alpha} := (\alpha_0, \alpha_1)^\top$ being the maximum likelihood estimate of $\alpha := (\alpha_0, \alpha_1)^\top$.

- (c) Modify the above obtained influence function of the resulting OLS estimator of the coefficient of `a` in the final model, in order to additionally account for the uncertainty due to estimating additional nuisance parameters, namely the parameters in the propensity score model (1). Try to use concise matrix notation if possible.
 - (d) Explain (in words) how you would use the obtained result to estimate the standard error of the coefficient of `a` in this analysis. Your answer can be brief, but it should be convincingly clear how you would do this in R.
 - (e) Do you expect the obtained standard error will agree with the one given by the standard output of the function `lm(y ~ a + ps)`? Why (not)?
2. (Exercise on the calculation of efficient influence curves) A mediation analysis aims to determine how much of the effect of an exposure A on an outcome Y is through an intermediate variable M . If the vector C contains all common causes of these 3 variables, then it can be shown that the so-called natural indirect effect (as an average difference) for a binary exposure can be identified as

$$\theta_{11}(P) - \theta_{10}(P),$$

where

$$\theta_{aa^*}(P) = \int E(Y|A = a, M = m, C = c) f(M = m|A = a^*, C = c) f(C = c) dm dc.$$

To derive the efficient influence function (canonical gradient) of $\theta_{aa^*}(P)$, take first the usual submodel $dP_{YAMC,t}(y, a, m, c) = t 1_{\tilde{y}=y, \tilde{a}=a, \tilde{m}=m, \tilde{c}=c} + (1-t) dP(y, a, m, c)$, where $1_{\tilde{y}=y, \tilde{a}=a, \tilde{m}=m, \tilde{c}=c}$ is the indicator function placing all the mass on the point $(\tilde{y}, \tilde{a}, \tilde{m}, \tilde{c})$, and show that

$$\begin{aligned} \frac{d}{dt}\bigg|_0 &= E_{P_t}[Y|A = 1, M = m, C = c] \\ &= \frac{1_{\tilde{a}=1, \tilde{m}=m, \tilde{c}=c}}{f_{AMC}(1, m, c)} (\tilde{y} - E[Y|A = 1, M = m, C = c]). \end{aligned}$$

Using this result, proceed to derive the efficient influence function of $\theta_{aa^*}(P)$.

3. (Exercise on the von Mises expansion) For a given $\tau \in (0, 1)$, let $\theta(P) = F^{-1}(\tau)$ be the τ th quantile of a continuously distributed random variable Y with distribution function $F(y) = P(Y \leq y)$, admitting density f and inverse F^{-1} . Recall that

$\theta(P)$ has efficient influence function $\phi_P(\tilde{y}) = \frac{\tau^{-1} \tilde{y} \leq \theta(P)}{f(\theta(P))}$. Assume that $\inf_y f(y) =: c > 0$ (this implies continuity of F and F^{-1}), and that $\sup_y f(y), \sup_y |f'(y)| < \infty$, where f' is the derivative of f , which is assumed to exist.

Let $\theta(\hat{P}_n) := \hat{F}^{-1}(\tau)$ and $\hat{f}$ be estimates of $\theta(P)$ and f , respectively, with both estimators using the same data corresponding to $\hat{P}_n$.

Let $\phi_{\hat{P}_n}(\tilde{y}) = \frac{\tau^{-1} \tilde{y} \leq \theta(\hat{P}_n)}{\hat{f}(\theta(\hat{P}_n))}$, and let $P_n := \frac{1}{n} \sum_{i=1}^n 1_{Y_i}$ be the empirical distribution generated by a sample $(Y_i)_{i \in [n]}$ which is independent from the data corresponding to $\hat{P}_n$.

- (a) Show that $E_P[\phi_P(Y)] = 0$.
- (b) Show that the plug-in bias

$$\begin{aligned} R_n &\equiv \theta(\hat{P}_n) - \theta(P) + P\phi_{\hat{P}_n} \\ &= \left(\theta(\hat{P}_n) - \theta(P)\right) \left(1 - \frac{f(\theta(\hat{P}_n))}{\hat{f}(\theta(\hat{P}_n))}\right) + O\left(\frac{(\theta(\hat{P}_n) - \theta(P))^2}{\hat{f}(\theta(\hat{P}_n))}\right), \end{aligned}$$

where a multiplicative constant $\sup_y |f'(y)|$ is hidden in O . (Hint: perform a Taylor expansion of $F(\theta(P))$ around the point $\theta(\hat{P}_n)$ with an appropriate representation of the remainder term.) Formulate the conditions — in addition to what is already assumed — under which $R_n = o_P(n^{-1/2})$.

- (c) Consider the empirical process term

$$\begin{aligned} T_n &\equiv \sqrt{n}(P_n - P)(\phi_{\hat{P}_n} - \phi_P) \\ &= \frac{1}{\sqrt{n}} \sum_{i=1}^n \left[\phi_{\hat{P}_n}(Y_i) - \phi_P(Y_i) - \int (\phi_{\hat{P}_n}(y) - \phi_P(y)) dP \right]. \end{aligned}$$

Show that if $\inf_y \hat{f}(y) =: c' > 0$, $\hat{f}(\theta(\hat{P}_n)) \xrightarrow{P} f(\theta(P))$, and $\theta(\hat{P}_n) \xrightarrow{P} \theta(P)$, then $T_n \xrightarrow{P} 0$. (Hint: you can make use of the continuous mapping theorem: if g is continuous at x and $X_n \xrightarrow{P} x$, then $g(X_n) \xrightarrow{P} g(x)$.)

- (d) As a conclusion of the above arguments, show that the first-step corrected estimator $\hat{\theta} := \theta(\hat{P}_n) + P_n\phi_{\hat{P}_n}$ is asymptotically efficient with variance $P\phi_P^2$, provided $\theta(\hat{P}_n)$ is root- n consistent and regularity conditions hold.